

Section 11.7 — Internal Rate of Return

Practice Worksheet — Tasks 1–10

Based exclusively on the concepts, formulas and worked example of Section 11.7: the net present value formula $A = a_0 + a_1/(1+r) + a_2/(1+r)^2 + \dots + a_n/(1+r)^n$, the internal rate of return equation (11.7.1), the substitution $s = (1+r)^{-1}$, the one-period formula $r = (b/a) - 1$, and the uniqueness/positivity theorem for $a_0 < 0$ with $a_1, \dots, a_n > 0$.

TASK 1

Difficulty: 1/5

A small bakery invests \$8,000 in a new commercial oven. The oven is expected to generate a single net return of \$9,600, paid in full at the end of one year.

Given:

Symbol	Value
a_0 (initial outlay)	-\$8,000
a_1 (return at end of Year 1)	\$9,600
n	1 year

Formula(s):

$$\text{For a one-year project: } -a + (1+r)^{-1} b = 0, \text{ so } r = (b/a) - 1$$

Step-by-Step Calculation:

Step 1: Identify $a = 8,000$ (amount invested) and $b = 9,600$ (amount returned).

Step 2: $r = (b/a) - 1 = (9,600 / 8,000) - 1 = 1.20 - 1 = 0.20 = 20\%$.

Step 3 (for later statements): NPV at rate r is $A = a_0 + a_1/(1+r)$.

Statements:

- A) The internal rate of return for this project is exactly 20%.
- B) At an interest rate of 15%, the net present value of this project is positive.
- C) At an interest rate of 25%, the net present value of this project is positive.
- D) If the return had instead been \$10,000 (outlay unchanged), the internal rate of return would exceed 24%.
- E) This project has a unique internal rate of return greater than -1.

Explanation of Statements:

A) $r = (9,600/8,000) - 1 = 0.20 = 20\%$, exactly as calculated above.

B) $NPV = -8,000 + 9,600/1.15 = -8,000 + 8,347.83 = \347.83 , which is positive (15% is below the 20% IRR, so NPV must be positive).

C) $NPV = -8,000 + 9,600/1.25 = -8,000 + 7,680.00 = -\320.00 , which is negative, not positive (25% is above the 20% IRR).

D) With $b = 10,000$: $r = (10,000/8,000) - 1 = 1.25 - 1 = 0.25 = 25\%$, which exceeds 24%.

E) The chapter's theorem states that if $a_0 < 0$ and all subsequent returns are positive, a unique internal rate of return $r^* > -1$ exists. Here $a_0 = -8,000 < 0$ and $a_1 = 9,600 > 0$, so the condition is satisfied.

A: TRUE | B: TRUE | C: FALSE | D: TRUE | E: TRUE

TASK 2

Difficulty: 2/5

A logistics company spends \$12,000 upgrading a delivery vehicle. The upgrade is expected to generate net returns of \$7,000 at the end of Year 1 and \$7,000 at the end of Year 2.

Given:

Symbol	Value
a_0	-\$12,000
a_1	\$7,000
a_2	\$7,000
n	2 years

Formula(s):

$$a_0 + a_1/(1+r) + a_2/(1+r)^2 = 0, \text{ substitute } s = (1+r)^{-1}$$

Step-by-Step Calculation:

Step 1: $-12,000 + 7,000/(1+r) + 7,000/(1+r)^2 = 0$.

Step 2: Let $s = (1+r)^{-1}$. Then $7,000s^2 + 7,000s - 12,000 = 0$. Dividing by 1,000: $7s^2 + 7s - 12 = 0$.

Step 3: Discriminant = $7^2 + 4(7)(12) = 49 + 336 = 385$. $\sqrt{385} \approx 19.6214$.

Step 4: $s = (-7 + 19.6214)/14 = 12.6214/14 \approx 0.90153$.

Step 5: $r = 1/s - 1 = 1/0.90153 - 1 \approx 0.10922 \approx 10.92\%$.

Statements:

A) The internal rate of return is approximately 10.92%.

B) At an interest rate of 8%, the net present value of this project is positive.

C) At an interest rate of 12%, the net present value of this project is positive.

D) If the Year 2 return had instead been \$8,000 (Year 1 unchanged at \$7,000), the internal rate of return would exceed 13%.

E) Doubling both returns to \$14,000 in Year 1 and \$14,000 in Year 2 (outlay unchanged at \$12,000) would result in an internal rate of return of approximately 21.84%.

Explanation of Statements:

A) As calculated above, $r \approx 10.92\%$.

B) $NPV = -12,000 + 7,000/1.08 + 7,000/1.1664 = -12,000 + 6,481.48 + 6,001.37 = \482.85 , positive (8% is below the IRR).

C) $NPV = -12,000 + 7,000/1.12 + 7,000/1.2544 = -12,000 + 6,250.00 + 5,580.36 = -\169.64 , negative, not positive (12% is above the IRR).

D) With $a_2 = 8,000$: $8,000s^2 + 7,000s - 12,000 = 0 \rightarrow 8s^2 + 7s - 12 = 0$. Discriminant = $49 + 384 = 433$, $\sqrt{433} \approx 20.809$. $s = (-7 + 20.809)/16 \approx 0.8630$, so $r = 1/0.8630 - 1 \approx 0.1587 = 15.87\%$, which exceeds 13%.

E) With $a_1 = a_2 = 14,000$: $14,000s^2 + 14,000s - 12,000 = 0 \rightarrow 7s^2 + 7s - 6 = 0$. Discriminant = $49 + 168 = 217$, $\sqrt{217} \approx 14.731$. $s = (-7 + 14.731)/14 \approx 0.5522$, so $r = 1/0.5522 - 1 \approx 0.8109 = 81.09\%$ - far more than double 10.92% (which would be $\approx 21.84\%$). Doubling the cash flows does not proportionally double the IRR.

A: TRUE | B: TRUE | C: FALSE | D: TRUE | E: FALSE

TASK 3

Difficulty: 2/5

A boutique invests \$20,000 in inventory and point-of-sale equipment. It expects net returns of \$9,000 at the end of Year 1 and \$15,000 at the end of Year 2.

Given:

Symbol	Value
a_0	-\$20,000
a_1	\$9,000
a_2	\$15,000
n	2 years

Formula(s):

$$a_0 + a_1/(1+r) + a_2/(1+r)^2 = 0, \quad s = (1+r)^{-1}$$

Step-by-Step Calculation:

Step 1: $15,000s^2 + 9,000s - 20,000 = 0$. Dividing by 1,000: $15s^2 + 9s - 20 = 0$.

Step 2: Discriminant = $9^2 + 4(15)(20) = 81 + 1,200 = 1,281$. $\sqrt{1,281} \approx 35.791$.

Step 3: $s = (-9 + 35.791)/30 \approx 0.89304$.

Step 4: $r = 1/0.89304 - 1 \approx 0.11978 \approx 11.98\%$.

Statements:

A) The internal rate of return is approximately 11.98%.

B) At a discount rate of 10%, the net present value of the project is positive.

C) At a discount rate of 14%, the net present value of the project is positive.

D) If the Year 1 return were \$9,000 higher, with Year 2 unchanged at \$15,000, the internal rate of return would exceed 30%.

E) The sum of all cash flows, $a_0 + a_1 + a_2$, equals \$4,000.

Explanation of Statements:

A) As calculated above, $r \approx 11.98\%$.

B) NPV = $-20,000 + 9,000/1.10 + 15,000/1.21 = -20,000 + 8,181.82 + 12,396.69 = \578.51 , positive (10% is below the IRR).

C) NPV = $-20,000 + 9,000/1.14 + 15,000/1.2996 = -20,000 + 7,894.74 + 11,542.01 = -\563.25 , negative, not positive (14% is above the IRR).

D) With $a_1 = 18,000$: $15,000s^2 + 18,000s - 20,000 = 0 \rightarrow 15s^2 + 18s - 20 = 0$. Discriminant = $324 + 1,200 = 1,524$, $\sqrt{1,524} \approx 39.038$. $s = (-18 + 39.038)/30 \approx 0.7013$, $r = 1/0.7013 - 1 \approx 0.4259 = 42.59\%$, which exceeds 30%.

E) $a_0 + a_1 + a_2 = -20,000 + 9,000 + 15,000 = \$4,000$, which is indeed positive, and this matches the chapter's stated result that the internal rate of return is positive whenever this sum is positive - consistent with the computed $r \approx 11.98\% > 0$.

A: TRUE | B: TRUE | C: FALSE | D: TRUE | E: TRUE

TASK 4

Difficulty: 3/5

A retailer is choosing between two one-year uses of surplus cash. Project X: invest \$15,000 now, receive \$17,250 in one year. Project Y: invest \$22,000 now, receive \$24,750 in one year.

Given:

Symbol	Value
Project X: a, b	\$15,000, \$17,250
Project Y: a, b	\$22,000, \$24,750
n	1 year (each project)

Formula(s):

$$r = (b/a) - 1 \quad (\text{one-year rate of return, applied to each project})$$

Step-by-Step Calculation:

Step 1: $r_X = (17,250/15,000) - 1 = 1.15 - 1 = 0.15 = 15\%$.

Step 2: $r_Y = (24,750/22,000) - 1 = 1.125 - 1 = 0.125 = 12.5\%$.

Statements:

A) The internal rate of return of Project X is exactly 15%.

B) The internal rate of return of Project Y is exactly 12.5%.

C) Based on the internal rate of return criterion, Project Y should be preferred over Project X.

D) At an interest rate of 11%, Project X has positive net present value while Project Y has negative net present value.

E) If Project Y's payoff had instead been \$25,000 (outlay unchanged at \$22,000), its internal rate of return would exceed that of Project X.

Explanation of Statements:

A) As calculated above, $r_X = 15\%$.

B) As calculated above, $r_Y = 12.5\%$.

C) The chapter's criterion is to prefer the project with the higher internal rate of return. Since 15% (X) is higher than 12.5% (Y), X should be preferred, not Y.

D) NPV_X at 11% = $-15,000 + 17,250/1.11 = -15,000 + 15,540.54 = \540.54 (positive). NPV_Y at 11% = $-22,000 + 24,750/1.11 = -22,000 + 22,297.30 = \297.30 , which is also positive, not negative, because 11% is still below Y's 12.5% IRR.

E) With $b = 25,000$: $r_Y = (25,000/22,000) - 1 \approx 0.1364 = 13.64\%$, which is still lower than Project X's 15%, so it would not exceed X's rate.

A: TRUE | B: TRUE | C: FALSE | D: FALSE | E: FALSE

TASK 5

Difficulty: 3/5

A manufacturing upgrade has an initial outlay of \$45,000. Installation disruption causes a net cash outflow of \$3,000 at the end of Year 1, followed by net returns of \$28,000 at the end of Year 2 and \$35,000 at the end of Year 3.

Given:

Symbol	Value
a_0	-\$45,000
a_1	-\$3,000
a_2	\$28,000
a_3	\$35,000
n	3 years

Formula(s):

$$A = a_0 + a_1/(1+r) + a_2/(1+r)^2 + a_3/(1+r)^3$$

Step-by-Step Calculation:

Step 1 (r = 8%): $A = -45,000 - 3,000/1.08 + 28,000/1.1664 + 35,000/1.259712 = -45,000 - 2,777.78 + 24,005.49 + 27,784.13 = \$4,011.84$.

Step 2 (r = 12%): $A = -45,000 - 3,000/1.12 + 28,000/1.2544 + 35,000/1.404928 = -45,000 - 2,678.57 + 22,321.43 + 24,912.31 = -\444.83 .

Step 3 (r = 15%): $A = -45,000 - 3,000/1.15 + 28,000/1.3225 + 35,000/1.520875 = -45,000 - 2,608.70 + 21,172.02 + 23,013.07 = -\$3,423.60$.

Statements:

- A) At r = 8%, the net present value of the project is approximately \$4,012.
- B) At r = 12%, the net present value of the project is positive.
- C) The internal rate of return of this project lies between 12% and 15%.
- D) At r = 15%, the net present value of the project is approximately -\$3,424.
- E) a_1 , a_2 , and a_3 are all positive.

Explanation of Statements:

A) As calculated in Step 1, $A \approx \$4,011.84$ at r = 8%, which rounds to approximately \$4,012.

B) As calculated in Step 2, $A \approx -\$444.83$ at r = 12%, which is negative, not positive.

C) The NPV is already negative at both r = 12% (-\$444.83) and r = 15% (-\$3,423.60), and it was positive at r = 8%. This means the rate where NPV crosses zero lies between 8% and 12%, not between 12% and 15%.

D) As calculated in Step 3, $A \approx -\$3,423.60$ at r = 15%, which rounds to approximately -\$3,424.

E) The theorem requires $a_0 < 0$ and all of $a_1, \dots, a_n$ to be strictly positive. Here $a_1 = -\$3,000$, which is negative, so this precondition is not met - the theorem's guarantee of a unique internal rate of return does not apply to this cash flow pattern.

A: TRUE | B: FALSE | C: FALSE | D: TRUE | E: FALSE

TASK 6

Difficulty: 4/5

A cafe chain invests \$34,000 in a new espresso machine line. Expected net returns are \$16,000 at the end of Year 1 and \$24,000 at the end of Year 2.

Given:

Symbol	Value
a_0	-\$34,000
a_1	\$16,000
a_2	\$24,000
n	2 years

Formula(s):

$$a_0 + a_1/(1+r) + a_2/(1+r)^2 = 0, \quad s = (1+r)^{-1}$$

Step-by-Step Calculation:

Step 1: $24,000s^2 + 16,000s - 34,000 = 0$. Dividing by 2,000: $12s^2 + 8s - 17 = 0$.

Step 2: Discriminant = $8^2 + 4(12)(17) = 64 + 816 = 880$. $\sqrt{880} \approx 29.665$.

Step 3: $s = (-8 + 29.665)/24 \approx 0.90270$.

Step 4: $r = 1/0.90270 - 1 \approx 0.10778 \approx 10.78\%$.

Statements:

- A) The internal rate of return is approximately 14.5%.
- B) At an interest rate of 9%, the net present value of the project is negative.
- C) At an interest rate of 13%, the net present value of the project is negative.
- D) If the Year 2 return had instead been \$20,000 (Year 1 unchanged at \$16,000), the internal rate of return would exceed the internal rate of return of the original project.
- E) Reducing the initial outlay to \$30,000 (returns unchanged at \$16,000 and \$24,000) would lower the internal rate of return.

Explanation of Statements:

- A) As calculated above, $r \approx 10.78\%$, not 14.5%.
- B) $NPV = -34,000 + 16,000/1.09 + 24,000/1.1881 = -34,000 + 14,678.90 + 20,200.32 = \879.22 , which is positive, not negative (9% is below the 10.78% IRR).
- C) $NPV = -34,000 + 16,000/1.13 + 24,000/1.2769 = -34,000 + 14,159.29 + 18,795.52 = -\$1,045.19$, which is negative, confirming the statement (13% is above the IRR).
- D) With $a_2 = 20,000$: $20,000s^2 + 16,000s - 34,000 = 0 \rightarrow 10s^2 + 8s - 17 = 0$. Discriminant $= 64 + 680 = 744$, $\sqrt{744} \approx 27.276$. $s = (-8 + 27.276)/20 \approx 0.9638$, $r = 1/0.9638 - 1 \approx 0.0375 = 3.75\%$, which is far lower than 10.78%, not higher.
- E) With $a_0 = -30,000$: $24,000s^2 + 16,000s - 30,000 = 0 \rightarrow 12s^2 + 8s - 15 = 0$. Discriminant $= 64 + 720 = 784 = 28^2$ exactly. $s = (-8 + 28)/24 = 20/24 \approx 0.83333$, $r = 1/0.83333 - 1 = 0.20 = 20\%$, which is much higher than the original 10.78% - a smaller outlay for the same returns raises the internal rate of return, it does not lower it.

A: FALSE | B: FALSE | C: TRUE | D: FALSE | E: FALSE

TASK 7

Difficulty: 4/5

A logistics firm spends \$40,000 automating a warehouse process. It expects net returns of \$22,000 at the end of Year 1 and \$27,600 at the end of Year 2.

Given:

Symbol	Value
a_0	-\$40,000
a_1	\$22,000
a_2	\$27,600
n	2 years

Formula(s):

$$A = a_0 + a_1/(1+r) + a_2/(1+r)^2$$

Step-by-Step Calculation:

Step 1: Test $r = 15\%$: $A = -40,000 + 22,000/1.15 + 27,600/1.3225 = -40,000 + 19,130.43 + 20,869.57 = \0.00 (to the nearest cent).

Step 2: Since $A = 0$ at $r = 15\%$, this confirms $r = 15\%$ is the internal rate of return for this project.

Statements:

- A) At $r = 15\%$, the net present value of the project is \$0 (to the nearest dollar).
- B) At an interest rate of 10%, the net present value of the project is positive.

- C) At an interest rate of 20%, the net present value of the project is negative.
 D) The sum of all the project's cash flows, $a_0 + a_1 + a_2$, equals \$9,600.
 E) This project has a unique internal rate of return greater than -1.

Explanation of Statements:

- A) As calculated above, $A = 0$ at $r = 15\%$, confirming this is the internal rate of return.
 B) $NPV = -40,000 + 22,000/1.10 + 27,600/1.21 = -40,000 + 20,000.00 + 22,809.92 = \$2,809.92$, positive (10% is below the 15% IRR).
 C) $NPV = -40,000 + 22,000/1.20 + 27,600/1.44 = -40,000 + 18,333.33 + 19,166.67 = -\$2,500.00$, negative (20% is above the 15% IRR).
 D) $a_0 + a_1 + a_2 = -40,000 + 22,000 + 27,600 = \$9,600$, which is positive, and this matches the chapter's result that the internal rate of return is positive whenever this sum is positive - consistent with the found $r = 15\% > 0$.
 E) The theorem's precondition ($a_0 < 0$, all subsequent $a_i > 0$) is satisfied exactly: $a_0 = -40,000 < 0$, $a_1 = 22,000 > 0$, $a_2 = 27,600 > 0$.

A: TRUE | B: TRUE | C: TRUE | D: TRUE | E: TRUE

TASK 8

Difficulty: 4/5

A property developer invests \$65,000 renovating a rental unit. Expected net rental income after expenses is \$34,000 at the end of Year 1 and \$42,000 at the end of Year 2.

Given:

Symbol	Value
a_0	-\$65,000
a_1	\$34,000
a_2	\$42,000
n	2 years

Formula(s):

$$a_0 + a_1/(1+r) + a_2/(1+r)^2 = 0, \quad s = (1+r)^{-1}$$

Step-by-Step Calculation:

- Step 1: $42,000s^2 + 34,000s - 65,000 = 0$. Dividing by 1,000: $42s^2 + 34s - 65 = 0$.
 Step 2: Discriminant $= 34^2 + 4(42)(65) = 1,156 + 10,920 = 12,076$. $\sqrt{12,076} \approx 109.891$.
 Step 3: $s = (-34 + 109.891)/84 \approx 0.90346$.
 Step 4: $r = 1/0.90346 - 1 \approx 0.10685 \approx 10.69\%$.

Statements:

- A) The internal rate of return is approximately 10.69%.
 B) At an interest rate of 9%, the net present value of the project is positive.
 C) At an interest rate of 12%, the net present value of the project is positive.
 D) Doubling both returns to \$68,000 (Year 1) and \$84,000 (Year 2), with the outlay unchanged at \$65,000, would more than double the internal rate of return.
 E) If the outlay were reduced to \$60,000 (returns unchanged), the internal rate of return would be lower than the internal rate of return of the original project.

Explanation of Statements:

A) As calculated above, $r \approx 10.69\%$.

B) $NPV = -65,000 + 34,000/1.09 + 42,000/1.1881 = -65,000 + 31,192.66 + 35,350.56 = \$1,543.22$, positive (9% is below the IRR).

C) $NPV = -65,000 + 34,000/1.12 + 42,000/1.2544 = -65,000 + 30,357.14 + 33,482.14 = -\$1,160.72$, negative, not positive (12% is above the IRR).

D) With $a_1 = 68,000$, $a_2 = 84,000$: $84,000s^2 + 68,000s - 65,000 = 0 \rightarrow 84s^2 + 68s - 65 = 0$. Discriminant = $4,624 + 21,840 = 26,464$, $\sqrt{26,464} \approx 162.678$. $s = (-68 + 162.678)/168 \approx 0.56356$, $r = 1/0.56356 - 1 \approx 0.7744 = 77.44\%$. Double the original rate would be about 21.36%, but the actual new rate (77.44%) is far higher than that - more than double.

E) With $a_0 = -60,000$: $42,000s^2 + 34,000s - 60,000 = 0 \rightarrow 42s^2 + 34s - 60 = 0$. Discriminant = $1,156 + 10,080 = 11,236 = 106^2$ exactly. $s = (-34 + 106)/84 = 72/84 \approx 0.85714$, $r = 1/0.85714 - 1 \approx 0.16667 = 16.67\%$, which is higher than 10.69%, not lower - a smaller outlay raises the internal rate of return.

A: TRUE | B: TRUE | C: FALSE | D: TRUE | E: FALSE

TASK 9

Difficulty: 5/5

A subscription software company spends \$50,000 building a product and is comparing two versions. Option 1 (full version): a steady net return of \$6,000 at the end of every year, indefinitely into the future. Option 2 (scaled-back version): the same \$50,000 outlay, but returns of only \$6,000 at the end of Year 1 and \$6,000 at the end of Year 2, with nothing paid afterward.

Given:

Symbol	Value
a_0 (both options)	-\$50,000
Option 1: a_i ($i = 1, 2, 3, \dots$)	\$6,000 each year, forever
Option 2: a_1, a_2	\$6,000, \$6,000 ($n = 2$)

Formula(s):

$$\text{Option 1 } (n \rightarrow \text{infinity}, a_i = a): a_0 + a/r = 0, \quad r = a / (-a_0)$$

$$\text{Option 2 } (n = 2): a_0 + a_1/(1+r) + a_2/(1+r)^2 = 0, \quad s = (1+r)^{-1}$$

Step-by-Step Calculation:

Step 1 (Option 1): As $n \rightarrow \text{infinity}$, $a_0 + a/r = 0$ gives $r = a/(-a_0) = 6,000/50,000 = 0.12 = 12\%$.

Step 2 (Option 2): $6,000s^2 + 6,000s - 50,000 = 0$. Dividing by 1,000: $6s^2 + 6s - 50 = 0$. Dividing by 2: $3s^2 + 3s - 25 = 0$.

Step 3: Discriminant = $3^2 + 4(3)(25) = 9 + 300 = 309$. $\sqrt{309} \approx 17.578$.

Step 4: The quadratic formula gives two roots for s : $s = (-3 - 17.578)/6 \approx -3.430$ and $s = (-3 + 17.578)/6 \approx 2.430$.

Step 5: Converting each root to $r = 1/s - 1$: the first root gives $r \approx -1.291$ (i.e. $r < -1$, outside the valid range), and the second gives $r \approx -0.588$ (i.e. $r > -1$, valid). So the unique valid internal rate of return for Option 2 is $r^* \approx -58.84\%$.

Statements:

A) The limiting internal rate of return of Option 1 is 12%.

B) The internal rate of return of Option 2 is approximately -58.84%.

C) Option 2 has a unique internal rate of return greater than -1.

D) The sum of Option 2's cash flows, $a_0 + a_1 + a_2$, equals -\$40,000.

E) If Option 2's Year 2 return were removed entirely, leaving only $a_0 = -\$50,000$ and $a_1 = \$6,000$, its internal rate of return would be even lower than Option 2's own internal rate of return.

Explanation of Statements:

A) As calculated in Step 1, $r = 6,000/50,000 = 0.12 = 12\%$.

B) As calculated in Steps 2-5, the valid root gives $r^* \approx -0.5884 = -58.84\%$.

C) The theorem only requires $a_0 < 0$ and all subsequent $a_i > 0$ to guarantee a unique $r^* > -1$; it does not require the quadratic to have only one mathematical root. Here one root ($r \approx -1.291$) falls outside the allowed range $r > -1$ and is discarded, leaving exactly one valid root - precisely what the theorem guarantees.

D) $a_0 + a_1 + a_2 = -50,000 + 6,000 + 6,000 = -\$38,000$, not $-\$40,000$.

E) With only $a_0 = -50,000$ and $a_1 = 6,000$: $r = (6,000/50,000) - 1 = 0.12 - 1 = -0.88 = -88\%$. Since -88% is lower (more negative) than -58.84% , the statement is correct.

A: TRUE | B: TRUE | C: TRUE | D: FALSE | E: TRUE

TASK 10

Difficulty: 5/5

A renewable energy cooperative is comparing two designs for a community solar project. Design A (two-year): invest \$120,000, with net returns of \$54,000 at the end of Year 1 and \$88,000 at the end of Year 2. Design B (simpler, one-year): invest \$70,000, with a single net return of \$81,200 at the end of Year 1.

Given:

Symbol	Value
Design A: a_0, a_1, a_2	-\$120,000, \$54,000, \$88,000
Design B: a, b	\$70,000, \$81,200
n	2 years (A) / 1 year (B)

Formula(s):

$$\text{Design A } (n = 2): a_0 + a_1/(1+r) + a_2/(1+r)^2 = 0, \quad s = (1+r)^{-1}$$

$$\text{Design B } (n = 1): r = (b/a) - 1$$

Step-by-Step Calculation:

Step 1 (Design A): $88,000s^2 + 54,000s - 120,000 = 0$. Dividing by 2,000: $44s^2 + 27s - 60 = 0$.

Step 2: Discriminant = $27^2 + 4(44)(60) = 729 + 10,560 = 11,289$. $\sqrt{11,289} \approx 106.250$.

Step 3: $s = (-27 + 106.250)/88 \approx 0.90057$, so $r_A = 1/0.90057 - 1 \approx 0.11041 \approx 11.04\%$.

Step 4 (Design B): $r_B = (81,200/70,000) - 1 = 1.16 - 1 = 0.16 = 16\%$.

Statements:

A) The internal rate of return of Design A is approximately 11.04%.

B) The internal rate of return of Design B is exactly 16%.

C) Based on the internal rate of return criterion, Design B should be preferred over Design A.

D) At a discount rate of 13%, Design A has negative net present value while Design B still has positive net present value.

E) If Design A's Year 1 return were \$10,000 lower, with Year 2 unchanged at \$88,000, its internal rate of return would still exceed Design B's 16%.

Explanation of Statements:

A) As calculated in Steps 1-3, $r_A \approx 11.04\%$.

B) As calculated in Step 4, $r_B = 16\%$ exactly.

C) The chapter's criterion is to prefer the project with the higher internal rate of return. Since 16% (Design B) exceeds 11.04% (Design A), Design B should indeed be preferred.

D) NPV_A at $13\% = -120,000 + 54,000/1.13 + 88,000/1.2769 = -120,000 + 47,787.61 + 68,916.91 = -\$3,295.48$, negative. NPV_B at $13\% = -70,000 + 81,200/1.13 = -70,000 + 71,858.41 = \$1,858.41$, positive. Both parts of the statement are correct.

E) With $a_1 = 44,000$: $88,000s^2 + 44,000s - 120,000 = 0 \rightarrow 22s^2 + 11s - 30 = 0$. Discriminant $= 121 + 2,640 = 2,761$, $\sqrt{2,761} \approx 52.545$. $s = (-11 + 52.545)/44 \approx 0.94420$, $r = 1/0.94420 - 1 \approx 0.05909 = 5.91\%$, which does not exceed Design B's 16% .

A: TRUE | B: TRUE | C: TRUE | D: TRUE | E: FALSE